VEC METRIC CARD

DES

DE gene recovery

de_score

HIGHER IS BETTER

T1 / T2 / T3

OFFICIAL WEIGHT CONTEXT

T1 25% | T2 expression group | T3 30%

THE QUESTION

Did the model recover the genes that truly changed?

WHAT IT REWARDS

Correct identity of response/developmental genes.

COMMON FAILURE

Mean shifts in the wrong genes, or missing the responsive set.

DEBUG NEXT

Compare predicted vs observed gene-level change against the same reference.

Community visual resource - MIT

VEC METRIC CARD

DCS

Change direction

de_direction

HIGHER IS BETTER T1 / T2 / T3

OFFICIAL WEIGHT CONTEXT

T1 25% | T2 expression group | T3 25%

THE QUESTION

Did genes move in the same ranked direction as reality?

WHAT IT REWARDS

Genome-wide directional agreement after controlling for reference expression.

COMMON FAILURE

Right magnitude but reversed/up-down response pattern.

DEBUG NEXT

Plot predicted vs true change vectors; inspect sign errors among strongest shifts.

Community visual resource - MIT

VEC METRIC CARD

MMD

Cell-state distribution

mmd_u

LOWER IS BETTER

T1 / T2 / T3

OFFICIAL WEIGHT CONTEXT

T1 30% | T2 cell-state group | T3 cell-state group

THE QUESTION

Is the generated population the same mixture of cell states?

WHAT IT REWARDS

Right states and proportions, not one average cell.

COMMON FAILURE

Population collapse, missing rare states, wrong mixture proportions.

DEBUG NEXT

Inspect cell-level diversity and embeddings; do not rely on pseudobulk alone.

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VEC METRIC CARD

CSS

Gene-gene co-variation

variogram

LOWER IS BETTER

T1 / T2 / T3

OFFICIAL WEIGHT CONTEXT

T1 20% | T2 cell-state group | T3 cell-state group

THE QUESTION

Are gene programs co-regulated together as in the target?

WHAT IT REWARDS

Correct pairwise co-expression structure across cells.

COMMON FAILURE

Correct marginals but shuffled gene relationships across cells.

DEBUG NEXT

Check covariance/co-expression patterns, not only per-gene distributions.

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VEC METRIC CARD

SDD

3D shape distance

d2_shape

LOWER IS BETTER

T2

OFFICIAL WEIGHT CONTEXT

1/3 of T2 shape+scale group (25%)

THE QUESTION

Does the predicted tissue have the right overall 3D shape?

WHAT IT REWARDS

Global geometric resemblance after allowed frame invariances.

COMMON FAILURE

Collapsed, distorted or anisotropic geometry.

DEBUG NEXT

Compare centered point-cloud scale and principal axes; visualize 3D cloud.

Community visual resource - MIT

VEC METRIC CARD

ODS

Occupancy Dice

occupancy_dice

HIGHER IS BETTER T2

OFFICIAL WEIGHT CONTEXT

1/3 of T2 shape+scale group (25%)

THE QUESTION

Do prediction and target occupy the same tissue volume?

WHAT IT REWARDS

Overlap of occupied spatial regions.

COMMON FAILURE

Wrong extent or holes even when gross shape looks plausible.

DEBUG NEXT

Voxelize/inspect coverage and spatial support, not only scatter appearance.

Community visual resource - MIT

VEC METRIC CARD

TSR

Tissue scale ratio

scale_log_ratio

ZERO IS IDEAL **T2**

OFFICIAL WEIGHT CONTEXT

1/3 of T2 shape+scale group (25%)

THE QUESTION

Is the predicted tissue at the right physical growth scale?

WHAT IT REWARDS

Matched global size; signed log ratio is scored by absolute distance from 0.

COMMON FAILURE

Correct-looking shape that is uniformly too large or too small.

DEBUG NEXT

Compare centered RMS radius / pairwise distance scale before changing topology.

Community visual resource - MIT

VEC METRIC CARD

NFS

Neighborhood feature similarity

neighborhood_mmd

LOWER IS BETTER T2

OFFICIAL WEIGHT CONTEXT

T2 local spatial organisation 25%

THE QUESTION

Are molecular cell states arranged in the right local neighborhoods?

WHAT IT REWARDS

Correct expression-location relationships locally.

COMMON FAILURE

Right global shape + right cells, but expression states scrambled across positions.

DEBUG NEXT

Inspect local neighbor composition and expression-position coupling.

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VEC METRIC CARD

SEVERITY

Response magnitude

severity_slope

ZERO IS IDEAL T3

OFFICIAL WEIGHT CONTEXT

T3 25%

THE QUESTION

Is the overall knockout response magnitude calibrated?

WHAT IT REWARDS

Predicted response amplitude matching the true perturbation amplitude.

COMMON FAILURE

Correct direction but systematically too weak or too strong.

DEBUG NEXT

Regress predicted change against true/reference-scale change; inspect slope bias.

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